Adaptive Evidence Acquisition for

Grounded Video Question Answering

A cost-aware evidence selection view of grounded VideoQA

Opening question

How much visual evidence is enough to answer a video question while staying temporally grounded?

Main claim

Answer accuracy, evidence cost, and temporal grounding must be evaluated together.

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Research Question 1

Q1. Do all video questions need the same amount of visual evidence?

Research Question 2

Q2. Can a compact learned policy preserve answer accuracy at lower cost?

Research Question 3

Q3. Is poor performance caused by weak evidence selection or by a weak answerer?

We answer these after defining the task, method, and evidence-budget experiments.

VideoQA evidence is distributed over time.

A correct option may be guessed from priors or broad scene context.

Fixed frame/clip budgets can waste evidence or miss the support moment.

Grounded VideoQA asks two things

1. Is the answer correct?
2. Does the selected evidence overlap the annotated support moment?

Why it matters

Grounding turns VideoQA from answer classification into evidence-supported reasoning.

Failure mode

correct but weakly grounded

Desired behavior

correct and grounded

Area	Examples	What they provide
VideoQA benchmarks	MovieQA, TGIF-QA, TVQA, NExT-QA	Temporal and causal reasoning
Grounded VideoQA	TVQA+, NExT-GQA	Answer must be supported by temporal evidence
Pretrained VLMs	CLIP, FrozenBiLM, SeViLA, Qwen2.5-VL	Strong visual-language reasoning
Retrieval-augmented video	VideoRAG-style selective context	Long-video evidence construction

Gap

Prior work often fixes the amount of context or reports answer quality without jointly charging evidence cost and measuring temporal support.

Grounded VideoQA example

$x = (v, q, A, y, T)$
video, question, answer options, label, support span

Evidence pool

$E(x) = \{e1, ..., eN\}$
frames and short temporal segments with cost and interval metadata

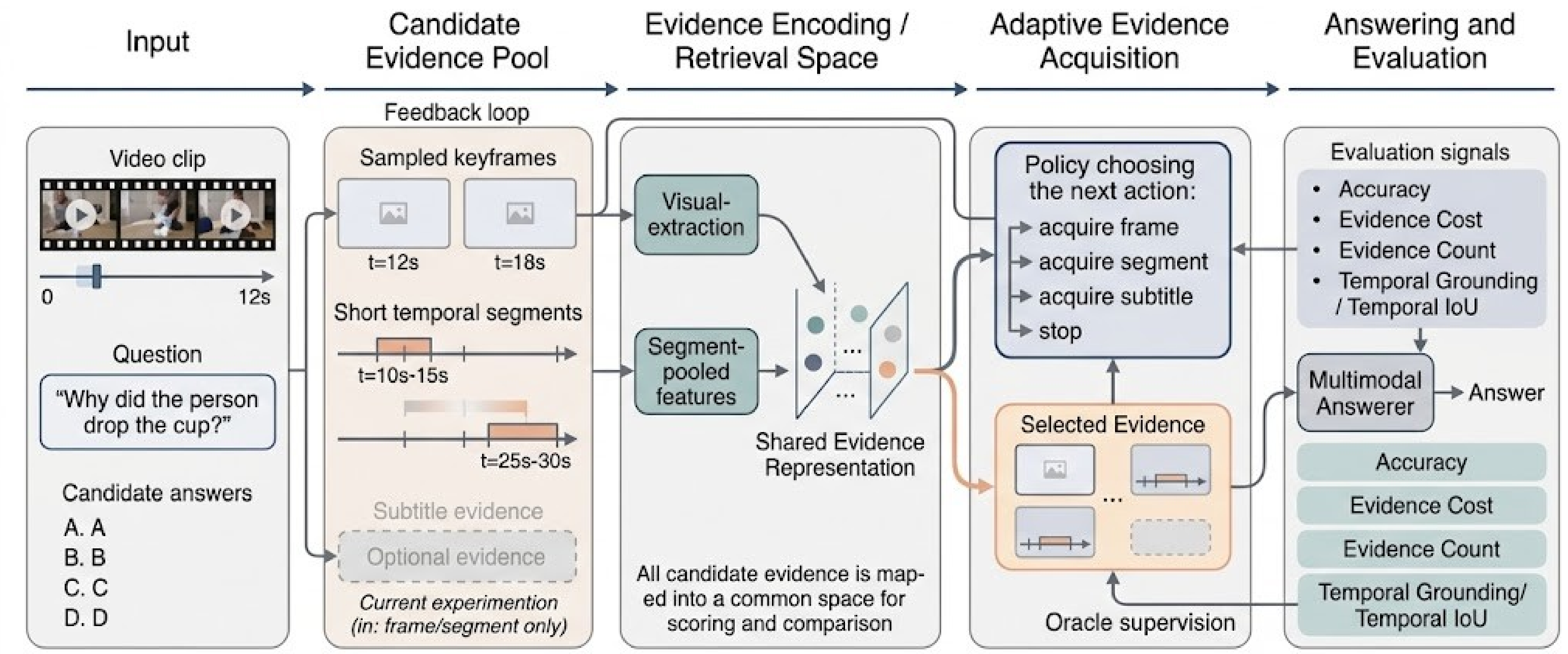
Policy

Selects S subset E(x)
C(S) = sum_i c_i

Answerer

Predicts answer from question, options, and selected evidence only.

Optimization intuition: maximize answer accuracy and grounding while minimizing acquired evidence cost.



Controlled answerer swap

Keep selected evidence fixed, then swap only the answerer to separate evidence quality from reasoning quality.

Policy	Selection	Count	Cost	Role
One segment	highest-scoring segment	1	1.500	Low-cost baseline
Learned two-item	sequential learned selection	2	2.795	Efficient middle point
Fixed 3+3	3 frames + 3 segments	6	7.500	High-coverage control
Fixed 6+6	6 frames + 6 segments	about 12	14.908	Large frozen-answerer control

Why fixed controls matter

They show what grounding improves when we spend much more visual context.

Why learned evidence matters

It tests whether compact evidence can keep most answer accuracy at lower cost.

Dataset

NExT-GQA
Validation: 3,358 examples
Test: 5,553 examples

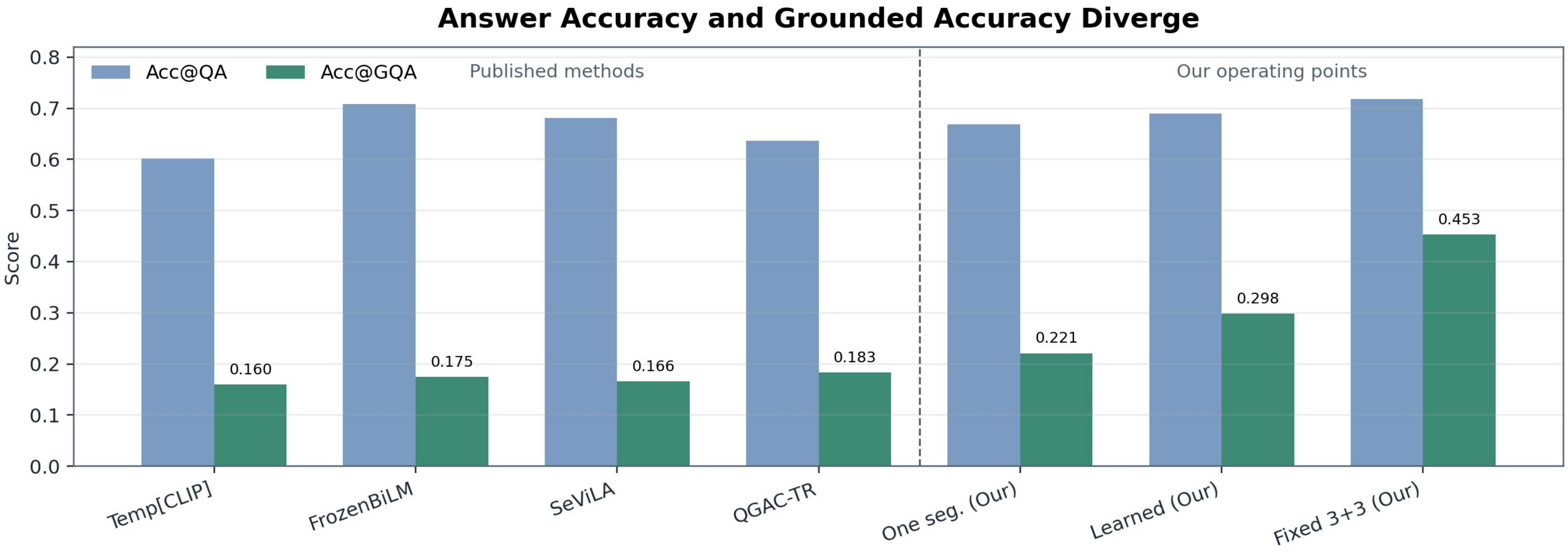
Answerers

Frozen CLIP-style scorer
Qwen2.5-VL-3B-Instruct

Protocol

Same selected evidence
Swap only final answerer

Metric	Meaning
Acc@QA	Whether the answer letter is correct
Cost / Count	Amount of visual evidence consumed
mIoP, IoP@0.5	How much selected evidence lies inside support span
mIoU, IoU@0.5	Stricter temporal overlap with support span
Acc@GQA	Correct answer plus IoP@0.5 grounding



Context

Prior methods show high Acc@QA but low Acc@GQA.

Our middle point

Learned evidence improves grounding at modest cost.

Our upper point

Fixed 3+3 gives the strongest grounding.

Method	Acc@QA	Cost	IoP@0.5	Acc@GQA
One segment	0.378	1.500	0.311	0.124
Learned two-item	0.381	2.795	0.410	0.164
Fixed 3+3	0.388	7.500	0.615	0.245
Fixed 6+6	0.394	14.908	0.787	0.316

Observation

Grounding improves as evidence budget increases, but answer accuracy remains near 0.38.

Conclusion

A weak answerer hides evidence-policy improvements in the answer metric.

Method	Acc@QA	Cost	IoP@0.5	Acc@GQA
Qwen one segment	0.668	1.500	0.311	0.221
Qwen learned two-item	0.689	2.795	0.410	0.298
Qwen fixed 3+3	0.718	7.500	0.615	0.453

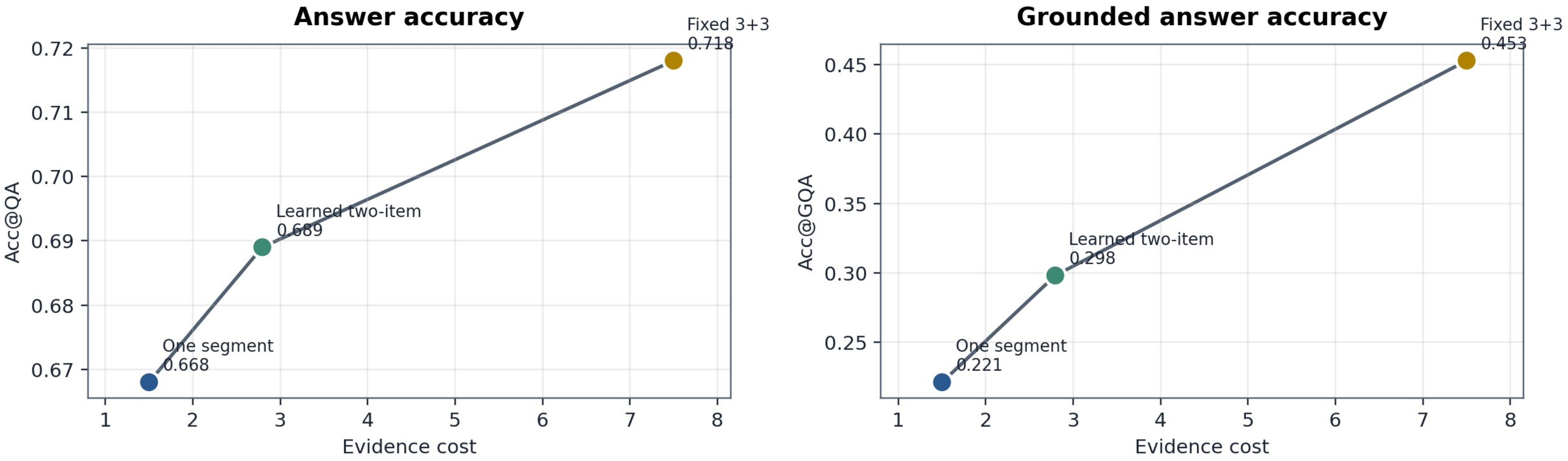
Answer retained
96%

Cost used
37%

Grounding retained
66%

Learned two-item evidence is not the maximum-grounding point; it is the efficient middle point.

Presentation Test-Set Summary: Accuracy-Cost-Grounding Tradeoff



Interpretation

The learned policy improves over one segment at modest cost; fixed 3+3 spends much more evidence to maximize grounding.

Case	What happens	Lesson
Correct and grounded	Evidence overlaps the support moment and Qwen predicts the gold answer.	Desired behavior
Correct, weakly grounded	Qwen predicts correctly but selected evidence has low temporal overlap.	Answer accuracy alone misleads
Grounded but wrong	Selected evidence overlaps support span, but answer reasoning fails.	Evidence and reasoning are separate

Takeaway

Grounded VideoQA needs both components: evidence policy must find the right moment, and the answerer must reason from it.

Q1. Same evidence budget for every question?

No. Evidence need varies; fixed context can be wasteful or weakly grounded.

Q2. Can compact learned evidence preserve accuracy?

Yes. Learned two-item evidence keeps about 96% of fixed 3+3 answer accuracy at 37% of the cost.

Q3. Evidence problem or answerer problem?

Both. The frozen answerer hides evidence gains; Qwen exposes the accuracy-cost-grounding frontier.

This answers the initial questions after the method and evidence-budget experiments.

Grounded VideoQA should be evaluated as evidence selection plus answer generation.

A weak answerer can hide better evidence selection.

A strong VLM makes compact evidence useful for answering.

Faithful temporal grounding still benefits from broader evidence coverage.

The result is a Pareto tradeoff, not a single winner.

One-line closing

Select enough evidence, not too much evidence, and verify that the answer is supported by the right moment.